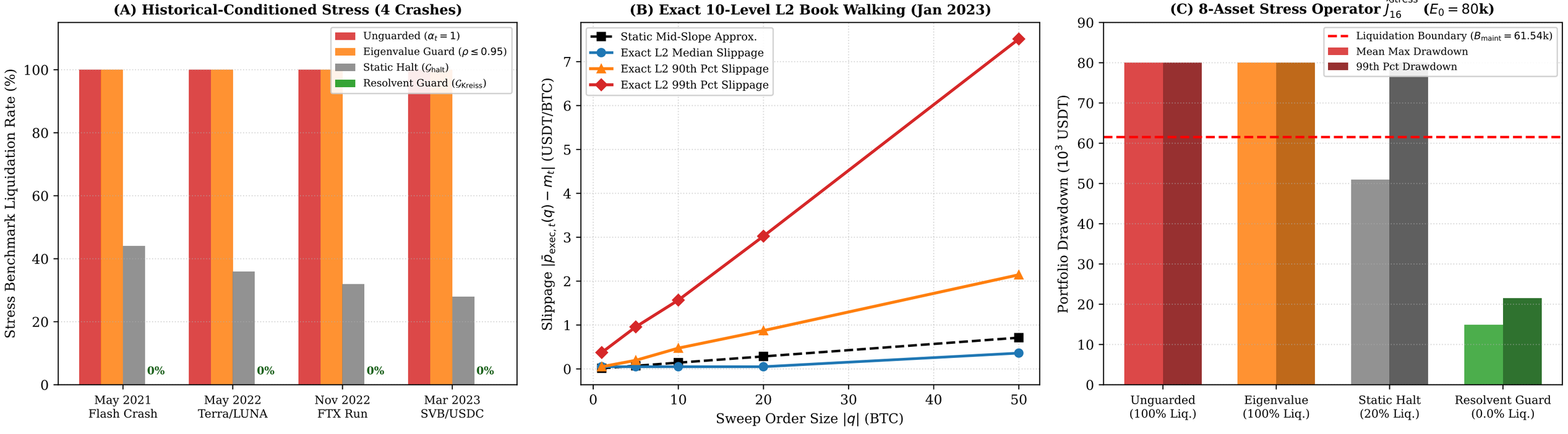

(A) Historical-Conditioned Stress (4 Crashes)
Stress Benchmark Liquidation Rate (%)
Unguarded ($\alpha_t = 1$)
Eigenvalue Guard ($\rho \le 0.95$)
Static Halt ($\mathcal{G}_{\text{halt}}$)
Resolvent Guard ($\mathcal{G}_{\text{Kreiss}}$)
0%
0%
0%
0%
May 2021 Flash Crash
May 2022 Terra/LUNA
Nov 2022 FTX Run
Mar 2023 SVB/USDC
(B) Exact 10-Level L2 Book Walking (Jan 2023)
Static Mid-Slope Approx.
Exact L2 Median Slippage
Exact L2 90th Pct Slippage
Exact L2 99th Pct Slippage
Slippage $|\bar{p}_{\text{exec},t}(q) - m_t|$ (USDT/BTC)
Sweep Order Size $|q|$ (BTC)
(C) 8-Asset Stress Operator $\hat{J}_{16}^{\text{stress}}$ ($E_0 = 80$k)
Liquidation Boundary ($B_{\text{maint}} = 61.54$k)
Mean Max Drawdown
99th Pct Drawdown
Portfolio Drawdown (10^3 USDT)
Unguarded (100% Liq.)
Eigenvalue (100% Liq.)
Static Halt (20% Liq.)
Resolvent Guard (0.0% Liq.)